

# Vasanth Avadhutha

## Software Engineer

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### SUMMARY

- Software Engineer with 4+ years of experience in developing robust, scalable applications for fintech and e-commerce sectors.
- Expertise in Java, Spring Boot, and React for full-stack development, with a proven track record of optimizing system performance and enhancing user experience.
- Proficient in designing and implementing microservices architectures, resulting in significant improvements in processing times and system scalability.
- Strong experience with AWS services including EC2, S3, and Lambda, for building efficient and cost-effective cloud-based solutions.
- Skilled in database management and optimization, working with MySQL, MongoDB, and implementing ORM technologies like Hibernate to enhance query performance.
- Adept at developing and integrating RESTful APIs, improving system interoperability and facilitating seamless data exchange between components.
- Strong background in data processing and analysis, utilizing technologies like Kafka for real-time data streaming and implementing machine learning models for predictive analytics.
- Experienced in Agile methodologies, CI/CD practices, and DevOps tools such as Jenkins, Docker, and Kubernetes for streamlined development and deployment processes.

### SKILLS

|                                 |                                                                                                                                                       |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming Languages:</b>   | Core Java, C, C++, SQL, C#                                                                                                                            |
| <b>Frameworks:</b>              | Struts, Hibernate, Spring Boot, Spring Batch, Spring Security, Spring AOP, Spring Core, Spring IOC, Angular 12, jQuery, Node.js, React.js, Express.js |
| <b>Web Technologies:</b>        | HTML, HTML5, CSS/CSS3, AJAX, jQuery, Bootstrap, XML, JSON, UI Material, SASS, Typescript, ES6, Redux, React Hooks, REST API, RESTful API              |
| <b>Java/J2EE Technologies:</b>  | Servlets, Spring, EJB, JPA, JTA, JDBC, JSP, JSTL, Webservices, Microservices, Spring MVC, Hibernate, ORM                                              |
| <b>Database:</b>                | Oracle SQL, MySQL, MongoDB, PostgreSQL, PL/SQL, DynamoDB                                                                                              |
| <b>Cloud Platforms:</b>         | Amazon Web Services                                                                                                                                   |
| <b>IDEs:</b>                    | Visual Studio Code, Eclipse IDE, IntelliJ IDEA, Spring Tool Suite (STS)                                                                               |
| <b>Web/Application Servers:</b> | Oracle WebLogic, IBM WebSphere, Apache Tomcat, JBoss                                                                                                  |
| <b>Testing Tools:</b>           | JUnit, Mockito, Selenium                                                                                                                              |
| <b>Build/ Other Tools:</b>      | Maven, Gradle, ANT, Jenkins, Docker, Kubernetes, SOAP/REST API, Postman, Jira, SonarQube, Kafka, Jupyter                                              |
| <b>Methodologies:</b>           | SDLC, Agile, Waterfall, SCRUM                                                                                                                         |
| <b>Version Control:</b>         | Git, GitHub, SVN, Bitbucket                                                                                                                           |
| <b>Operating System:</b>        | Windows, Linux, Mac OS                                                                                                                                |

### EXPERIENCE

Software Engineer | **Mastercard** – MO

June 2023 – Present

- **Developed dynamic and responsive web interfaces** using **React.js**, incorporating modern web technologies like **HTML5**, **CSS3**, and **JavaScript** for building interactive components and seamless user experiences.
- **Developed transaction management functionality**, allowing users to transfer funds between accounts, view balances, and track transaction history in real time, improving **transaction processing speed by 30%** and enhancing **user engagement by 25%** due to real-time updates and intuitive interface.
- **Integrated Spring Security** to implement secure login and authentication mechanisms using **JWT (JSON Web Tokens)** for stateless authentication across the web application, reducing **login time by 30%** and improving overall security by **40%** through more robust session management and token-based validation.
- **Utilized Java Persistence API (JPA) and Hibernate ORM** to map Java objects to relational database tables in **MySQL**, enabling seamless interaction with the database and improving **development speed by 25%** and **database query performance by 40%** due to optimized SQL generation and caching mechanisms.

- Implemented a global exception handling mechanism using **@ControllerAdvice** to capture and manage application errors such as invalid user inputs or database constraints violations, returning appropriate HTTP response codes and messages.
- Used **Java's concurrency utilities** (Executor Service, Synchronized) to handle concurrent requests, ensuring that multiple transactions (e.g., transfers) could be processed simultaneously without data inconsistencies.
- Designed and implemented **RESTful APIs** using **Spring Boot** to handle various banking functionalities, including **user registration, fund transfers, balance retrieval, and transaction history**.
- Integrated **Amazon S3** to store and serve static files such as **profile pictures, transaction receipts, and logs** in a highly available and durable storage solution. Configured S3 bucket policies to ensure secure access to stored files, and used **presigned URLs** for temporary, secure access to files when needed.
- Implemented **Apache Kafka** to enable real-time data streaming and event-driven architecture for processing banking transactions (fund transfers, transaction status updates).
- Actively participated in **Agile ceremonies** such as sprint planning and daily standups. Collaborated closely with product owners, developers, and QA teams to ensure timely and high-quality deliverables.

#### Java Full Stack Developer | **Capgemini, India**

**Oct 2018 – Dec 2021**

- Designed and implemented a **microservices-based web application** using **Spring Boot** and **Java 8** to streamline **customer onboarding** and **transaction processing**, ensuring a scalable and maintainable architecture.
- Developed **MongoDB** to handle user-generated content, such as transaction logs, customer profiles, and account settings, ensuring efficient data storage and retrieval for large-scale applications.
- Created **RESTful web services** with **Spring Boot** to expose APIs for critical features like user authentication, transaction tracking, and real-time updates, improving system interoperability and extensibility.
- Used **Angular Reactive Forms** for building **dynamic, form-based UI components** like transaction forms, ensuring real-time validation and improving data accuracy by **30%** during fund transfers and user input.
- Executed secure communication using **Java's SSL/TLS** libraries to encrypt sensitive user and transaction data between the client and server, improving data security and reducing the risk of data breaches by **60%**.
- Designed **consistent UI components** such as navigation bars, buttons, forms, and modals using **Bootstrap's pre-built components**, reducing development time and improving consistency across the application.
- Made **Model-View-Controller (MVC) architecture** using **Apache Struts** to separate concerns and structure the web application, ensuring maintainable and scalable code.
- Established **multithreading** in Java to process concurrent transactions in real-time, improving the transaction throughput and ensuring **high availability** for banking services during peak traffic.
- Integrated **JDBC with Java services** to process real-time data operations such as retrieving account balances, transaction history, and performing balance updates, allowing users to view updated information instantly.
- Implemented **caching mechanisms** using **Redis** to reduce database load for frequently accessed data (e.g., account balances, transaction history), improving response times by **50%** for end users.
- Developed core functionality using **Java Servlets** to handle HTTP requests and responses, enabling the processing of user transactions, account management, and fund transfers in the banking application, improving transaction processing speed and efficiency by **35%**.
- Combined **JavaScript with frontend libraries** such as **jQuery** to simplify DOM manipulation, event handling, and AJAX calls, reducing development time and ensuring smooth user interactions.
- Containerized the **full-stack banking application** using **Docker** to encapsulate the backend, frontend, and database services into separate containers, ensuring consistent environments across development, testing, and production stages.
- Configured **CI/CD pipelines** using **Jenkins**, automating the entire process from code commit to production deployment for the banking application, reducing manual intervention and deployment time by **40%**.
- Used **Mockito** to simulate various user inputs and scenarios (valid fund transfer, invalid account number) in service methods, ensuring that business logic was validated and edge cases were tested thoroughly.

## EDUCATION

**Master of Science in Computer Science** – Saint Louis University St Louis MO, USA

**Bachelor of Technology in Computer science and Engineering** – SR ENGINEERING COLLEGE Hasanparthy, India